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FedCART: Tackling Long-Tailed Distributions in Federated Adversarial Training via Classifier Refinement

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Abstract

Growing privacy and security demands in the real world have spurred interest in adversarially robust Federated Learning (FL). While Adversarial Training (AT) is a well-established defense in centralized learning, its extension to the federated setting, known as Federated Adversarial Training (FAT), faces significant challenges due to data heterogeneity across clients. Existing FAT methods have made significant contributions, but they typically assume a balanced global data distribution, an assumption that rarely holds true in practice due to the prevalence of long-tailed distributions. This work first identifies and diagnoses the severe performance degradation of FAT under long-tailed data, attributing it to skewed feature representations and impaired classifier discriminability. To address this, we propose FedCART, a novel FAT framework that decouples the model into a shared feature extractor and a dual-classifier structure. On the client side, a representation-alignment loss enhances adversarial robustness, while gradient-based class prototypes are extracted for classifier calibration. On the server side, models and prototype sets are aggregated to synthesize balanced virtual features, enabling the re-training of an auxiliary classifier to mitigate long-tailed bias. Extensive experiments demonstrate that FedCART significantly improves both accuracy and robustness, outperforming state-of-the-art FAT methods. To the best of our knowledge, this is the first work to systematically investigate and address FAT under long-tailed distributions, representing a significant step toward practical adversarial robustness in FL.

1. Introduction

Federated Learning (FL) [21] has emerged as a promising paradigm for collaborative model training while preserving data privacy. However, recent studies reveal that FL remains vulnerable to adversarial attacks [5, 6, 30] similar to those in traditional machine learning. Specifically, during inference, an adversary can craft human-imperceptible perturbations to benign/natural inputs, generating adversarial samples that successfully fool the global model with high probability.

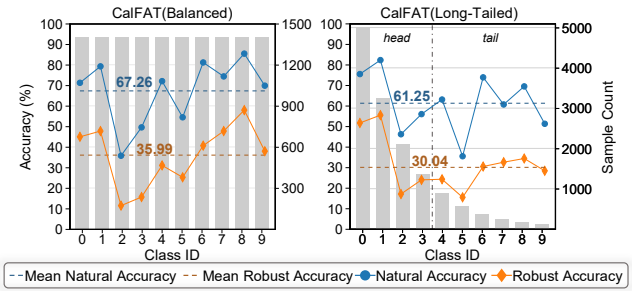


Figure 1. Performance of CalFAT [3] under **Left:** class-balanced and **Right:** long-tailed (i.e., class-imbalanced) distribution. We define head classes as those with over 1000 samples in the long-tailed partition (classes 0-3), with the remainder designated as tail classes. Our results reveal a significant predictive bias toward these head classes. This demonstrates the detrimental effect of long-tailed data on model performance and robustness.

In conventional machine learning, adversarial training (AT) [20, 47] is among the most effective defenses against adversarial attacks. However, implementing AT in a federated setting, known as Federated Adversarial Training (FAT) [50], presents unique challenges due to the Non-Independent and Identically Distributed (Non-IID) [10, 15, 17] nature of data across clients. While recent efforts [3, 4, 8, 24, 27, 44, 49] like CalFAT [3] have begun exploring FAT under Non-IID conditions, they predominantly operate under a closed-world assumption. This assumes that all local data are drawn from a global, labeled, and class-balanced dataset, a condition that rarely holds in practice. In contrast, real-world, open-world scenarios often feature data with long-tailed distributions [14, 46].

As our first contribution, we investigate the performance of CalFAT under long-tailed data distributions. As illustrated in Fig. 1, a shift from a class-balanced to a long-tailed (i.e., class-imbalanced) global distribution leads to a marked

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decline in both overall natural and robust accuracies[1]. We define head classes as those with over 1000 samples in the long-tailed partition (classes 0-3), with the remainder designated as tail classes. Our results reveal a significant predictive bias toward these head classes. This demonstrates the detrimental effect of long-tailed data on model performance and robustness. We attribute this performance degradation to the confluence of inter-client class imbalance and the global long-tailed distribution in FAT, which jointly corrupts feature representations and impairs the classifier's discriminative power. A detailed analysis is provided in Sec. 3.3.

Our second contribution: Motivated by these observations, we propose FedCART, a novel client-server collaborative framework for FAT. To the best of our knowledge, FedCART is the first FAT method specifically designed to handle long-tailed data distributions. The core innovation is the decoupling of the global model into a shared feature extractor and a dual-classifier structure. This allows FedCART to strategically separate the challenges of robust feature learning and unbiased classification. On the client side, FedCART employs a contrastive learning-based feature alignment mechanism. This adaptively enhances the consistency between natural and adversarial samples in the feature space, which stabilizes local training and bolsters model robustness against attacks. On the server side, the framework leverages a gradient prototyping mechanism to perform virtual feature alignment and classifier retraining. By aggregating class gradient prototypes from clients, the server generates a balanced virtual feature set. An auxiliary classifier is then retrained on this set, effectively mitigating the classification bias inherent in global long-tailed distributions. Crucially, this entire process adheres to the privacy principles of FL, as it requires no sharing of raw data.

As our third contribution, we conduct comprehensive and in-depth experimental validation. We perform extensive experiments on multiple standard datasets and under different settings with varying degrees of long tails and data heterogeneity. The results consistently demonstrate that FedCART significantly outperforms state-of-the-art FAT methods in both natural accuracy and adversarial robustness, particularly exhibiting superior advantages in preserving tail class performance and handling highly Non-IID data. Detailed ablation experiments also confirm the necessity and effectiveness of each core component of the framework. We envision this work represents an important step toward building robust and equitable FL systems for complex real-world data distributions.

The main contributions of this paper are summarized as follows:

- **Problem Identification.** We reveal and diagnose the severe performance degradation of FAT on long-tailed data. We attribute that the root cause is the classifier's failure under skewed and distorted feature representations due to adversarial attacks and the scarcity of tail class.
- **Novel Framework.** We propose FedCART, a client-server collaborative FAT framework and, to our knowledge, the first specifically designed for long-tailed distributions. Its core innovation is decoupling the model into a shared feature extractor and a dual-classifier. This structure strategically separates the challenges: clients perform contrastive feature alignment for robustness, while the server uses gradient prototyping to refine the classifier and eliminate bias, all without violating FL privacy.
- **Extensive Validation.** Through comprehensive experiments on multiple datasets under varying long-tailed and Non-IID settings, we demonstrate that FedCART significantly outperforms state-of-the-art FAT methods in both natural accuracy and adversarial robustness. Ablation studies confirm the necessity and effectiveness of each core component.

[1]Following [3], natural accuracy means the test accuracy for benign/natural inputs, and robust accuracy means the test accuracy for inputs under adversarial attack.

2. Related Work

2.1. Federated Long-Tailed Learning

FL has attracted significant attention since the introduction of FedAvg [21]. Nevertheless, severe Non-IID characteristics among clients degrade the performance of federated model [15, 17, 45] and obstacles to real-world application [39]. To address this, numerous methods have been proposed, including prototype-based [31, 43] and classifier improvement-based approaches [16, 18]. Moreover, long-tailed data are pervasive in practice, yet most FL methods assume balanced global distributions [14, 28, 29, 48]. Severe class imbalance compounded with Non-IID among client splits amplifies head-class bias, degrades tail representations, and impairs model performance [11, 22, 48]. While initial efforts like reweighting-based loss designs [26, 33, 37] have been proposed, they often struggle with inaccurate global distribution estimation and tend to overfit tail classes. Many other approaches leverage prototype or gradient sharing for classifier calibration [9, 28, 29, 38, 48]. Although existing methods have made promising progress in mitigating the impact of long-tailed distributions, they primarily operate in benign settings, and their adversarial robustness is unexplored. Our work bridges this gap.

2.2. Federated Adversarial Training

Adversarial attacks [2, 5, 6, 20] can severely degrade model performance, making robustness a critical concern. Although AT has demonstrated effectiveness in centralized learning [20, 34, 35, 40–42], its direct extension as FAT faces significant challenges due to the data heterogeneity among clients [44]. FedFAT [50] first introduced FAT in the

context of FL. Subsequent studies have further advanced FAT by addressing a variety of key FL-specific challenges, such as communication constraints [27], or limited computational resources for clients [8, 32]. To address data heterogeneity, prior efforts have improved decision boundaries [4, 44], adversarial sample generation [19], and server aggregation mechanisms [49]. CalFAT [3] adapts both loss functions and the generation of adversarial samples using local label distributions. Other approaches leverage self-distillation for feature enhancement [24] or utilize feature statistics for classifier calibration [25]. Despite these advances, most existing methods primarily target generic Non-IID settings and overlook the practical challenge posed by long-tailed data distributions. In such scenarios, these approaches often underperform due to the scarcity of samples from rare classes. However, our work considers a practical and critical problem: FAT under the long-tailed plight, with the goal of rectifying biased global optimization. Inspired by [12, 18, 29], we propose FedCART, a FAT approach explicitly designed for long-tailed data, to address this issue.

3. Preliminaries

3.1. Federated Learning

In FL, a server coordinates K clients, each holding a private dataset $\mathcal{D}_k$ of size N_k. The objective is to collaboratively learn a shared model $\mathbf{w}$ over $\mathcal{D} = \bigcup_{k=1}^{K} \mathcal{D}_k$ in T rounds without sharing raw data. For $\mathcal{D}$ comprising $|C|$ classes, let $n_k^{(c)}$ denote the number of samples of class c on the client k, and $n^{(c)} = \sum_{k=1}^{K} n_k^{(c)}$. We consider $\mathcal{D}$ follows a long-tailed distribution, where satisfies $n^{(1)} \gg n^{(2)} \gg \cdots \gg n^{(|C|)}$, resulting in a few head classes and many tail classes. The model $\mathbf{w} = (\mathbf{r}, \mathbf{h})$ consists of a feature extractor $\mathbf{r}$ and a classifier $\mathbf{h}$. Given an input x, we obtain feature $z = \mathbf{r}(x)$, logits $s = \mathbf{h}(z)$, and prediction $\hat{y}$.

At round t, a subset $A^{(t)}$ of clients performs local training on $\mathbf{w}_k^{(t)}$ and uploads their updated $\mathbf{w}_k^{(t+1)}$. The server aggregates them via weighted average like FedAvg [21]:

$$\mathbf{w}^{(t+1)} \leftarrow \sum_{k \in A^{(t)}} \frac{N_k}{\sum_{k' \in A^{(t)}} N_{k'}} \mathbf{w}_k^{(t+1)}. \tag{1}$$

The aggregated model is then distributed to clients for the next round.

3.2. Adversarial Training

Adversarial training (AT) [6, 20] has been extensively studied as one of the most effective defense mechanisms to enhance adversarial robustness, by minimizing the loss of samples with crafted perturbations. Specifically, the objective of AT [20] can be expressed as a multi-step min-max process:

$$\min_{\mathbf{w}} \max_{\|\delta\|_p \leq \epsilon} \left(\mathcal{L}_{\text{CE}}(\mathbf{w}; x + \delta, y)\right) \tag{2}$$

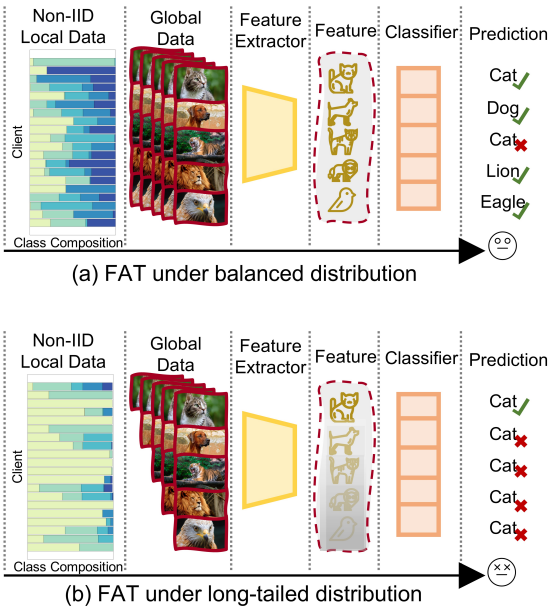


Figure 2. The illustration of FAT under (a) balanced and (b) long-tailed data. Existing FAT methods only consider the balanced global data but overlook the unique challenge induced by long-tailed data. Long-tailed data intensifies inter-client inconsistency during training by concentrating tail classes on a more limited set of clients, which makes the features of tail classes extremely ambiguous. For each client, the scarcity of tail-class samples yields a biased local model and skewed features, ultimately confusing the classifier.

where $\mathcal{L}_{\text{CE}}(\cdot)$ denotes the cross-entropy (CE) loss. The adversarial perturbation δ is generated within a bounded perturbation budget ϵ and $\|\cdot\|_p$ denotes the p-norm constraint. And PGD [20] is widely adopted for generating adversarial samples. Let $x' = x + \delta$ denote the adversarial sample, its feature, logit, and prediction are denoted as z', s', and $\hat{y}'$, respectively, in the following discussion.

3.3. Motivation

As discussed in the previous section, both the natural and robust accuracies of CalFAT degrade under long-tailed distributions. Compared with balanced data, long-tailed data pose a unique challenge, as illustrated in Fig. 2.

Under the long-tailed distribution, tail classes become even scarcer on clients. In Non-IID settings, data distributions on client deviate further, with tail classes concentrated on only a few clients. This imbalance causes local models to learn divergent features, leading to inconsistent local updates. Consequently, the aggregated global model exhibits a skewed and chaotic feature space. Within each client, class imbalance biases the model toward dominant classes. Moreover, adversarial samples differ substantially from natural samples, which seriously confuses the classifier.

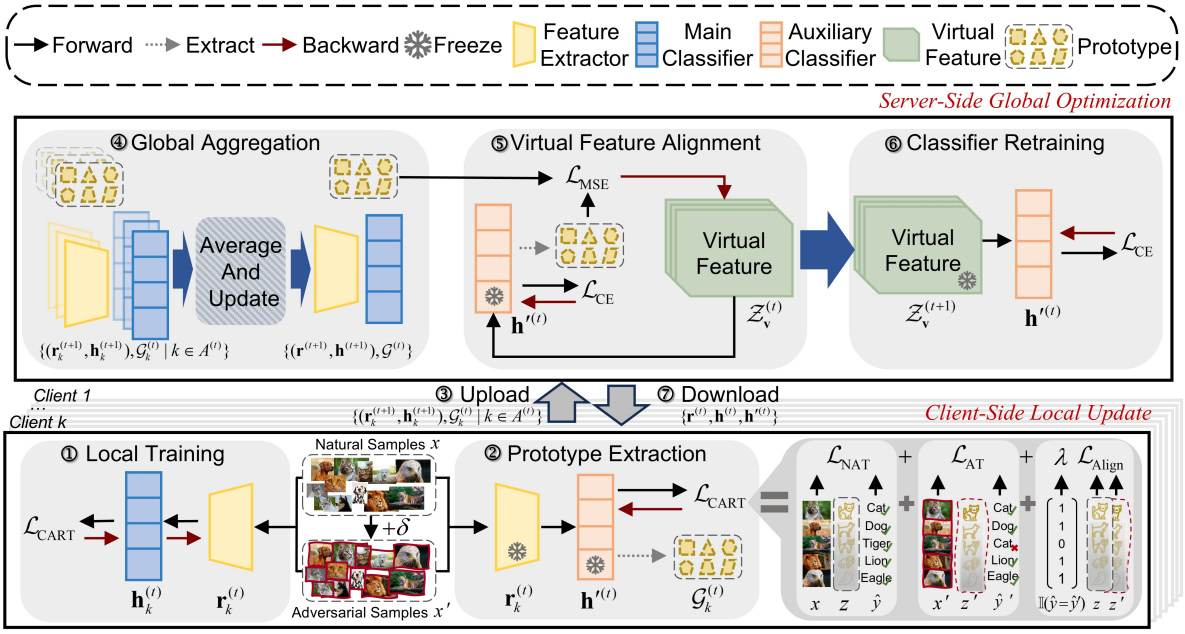


Figure 3. The framework of FedCART. In each communication round t, clients conduct local training following $\mathcal{L}_{CART}$ to enhance the robustness against attacks, and extract the gradient-baesd prototypes by the auxiliary classifier. The server aggregates the models and prototypes from active clients, then constructs a set of balanced virtual features based on aggregated prototypes for the re-training of the auxiliary classifier. Finally, the server distributes the model to the clients participating in the next round.

We argue that the combined effect yields highly distorted feature representations, severely impairing classifier performance. To mitigate this, we propose FedCART, which aligns natural and adversarial features on clients and refines the classifier on the server, jointly improving discriminability and robustness.

4. Proposed Method

4.1. Framework

To tackle the challenges described in Sec. 3.3, we introduce a simple yet effective framework, FedCART, which decomposes the global model into a shared feature extractor $\mathbf{r}$ and a dual-classifier structure, the main classifier $\mathbf{h}$ and the auxiliary one $\mathbf{h}'$. As illustrated in Fig. 3, FedCART consists of two core components: (i) client-side local updates and (ii) server-side global optimization. The central idea is to align feature representations of natural and adversarial samples on clients and refine the classifier via a set of balanced virtual features on the server, thereby mitigating the performance degradation induced by long-tailed, heterogeneous data in FAT.

Client-side Local Update.(Sec. 4.2) In round t, each active client $k \in A^{(t)}$ downloads the global model from the server, and updates it for E epochs following a contrastive learning-based feature-alignment loss $\mathcal{L}_{\text{CART}}$ to enhance the robustness against attacks. In addition, each client extracts a set of gradient-based class prototypes for calibration of the auxiliary classifier. Finally, clients upload the updated model and class prototypes to the server.

Server-side Global Optimization.(Sec. 4.3) After the server aggregating models and prototypes from clients. The server then generates a set of virtual features, whose gradients for the auxiliary classifier align with the aggregated prototypes, to refine. This process effectively calibrates the bias induced by the long-tailed and heterogeneous data, thereby improving the discriminability. The refined global model is then distributed to participants in the next round.

Inference Stage. When training is completed in round T, we deploy the auxiliary classifier $\mathbf{h}'^{(T)}$ with the final extractor $\mathbf{r}^{(T)}$ for inference, discarding the main classifier $\mathbf{h}^{(T)}$.

4.2. Client-Side Local Update

For each client k, the local update process consists of two components: (i) local training following $\mathcal{L}_{\text{CART}}$ on the private dataset $\mathcal{D}_k$, and (ii) extraction of gradient-based class prototypes $\mathcal{G}_k^{(t)}$. The procedure of the client-side local update is summarized in Algorithm 1 in detail.

Local Training. In round t, upon receiving the global model with the shared feature extractor $\mathbf{r}^{(t)}$, the main classifier $\mathbf{h}^{(t)}$, and the auxiliary classifier $\mathbf{h}'^{(t)}$ from the server, the client k initializes its local model with $\mathbf{w}_k^{(t)} = (\mathbf{r}^{(t)}, \mathbf{h}^{(t)})$,

then optimizes it following the proposed loss function $\mathcal{L}_{\text{CART}}$, which considers the natural and adversarial samples with a representation-alignment mechanism:

$$\mathcal{L}_{\text{CART}} = \mathcal{L}_{\text{NAT}} + \mathcal{L}_{\text{AT}} + \gamma \mathcal{L}_{\text{Align}}. \tag{3}$$

Here, $\mathcal{L}_{\text{NAT}} = \mathcal{L}_{\text{CE}}(\mathbf{w}_k^{(t)}; x, y)$ and $\mathcal{L}_{\text{AT}} = \mathcal{L}_{\text{CE}}(\mathbf{w}_k^{(t)}; x', y)$ are the losses for natural and adversarial samples, respectively.

To enforce consistency between the feature representations z and z' of natural and adversarial samples, we propose a contrastive learning-based feature alignment term $\mathcal{L}_{\text{Align}}$:

$$\mathcal{L}_{\text{Align}} = \sum_{i=1}^{2N_k} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i \cdot z_p'/\tau)}{\sum_{j \in Q(i)} \exp(z_i \cdot z_j'/\tau)}, \tag{4}$$

where $P(i)$ indexes samples that share the same label as y_i, and $Q(i)$ includes all other samples. τ is a temperature factor.

To adaptively adjust alignment strength, we introduce a scaling factor γ based on the consistency of the prediction:

$$\gamma = \iota \left(1 - \frac{\sum_{i=1}^{N_k} \mathbb{I}(\hat{y}_i = \hat{y}_i')}{N_k}\right), \tag{5}$$

where $\mathbb{I}(\cdot)$ is the indicator function and ι is a scaling constant. Intuitively, γ strengthens alignment when natural and adversarial predictions diverge, while avoiding unnecessary penalties.

The local model is then optimized via stochastic gradient descent following the proposed loss $\mathcal{L}_{\text{CART}}$ for E epochs, integrating both adversarial robustness and feature consistency.

Extraction of Gradient-based Class Prototypes. Client k initializes its local model as $\mathbf{w}_k'^{(t)} = (\mathbf{r}^{(t)}, \mathbf{h}'^{(t)})$ and extracts the gradient-based prototypes $\mathbf{g}_{k,c}^{(t)}$ of class c by the auxiliary classifier $\mathbf{h}'^{(t)}$:

$$\mathbf{g}_{k,c}^{(t)} = \frac{1}{n_k^{(c)}} \sum_{i=1}^{n_k^{(c)}} \nabla_{\mathbf{h}'} \mathcal{L}_{\text{CART}} \left(\mathbf{w}_k'^{(t)}; x_i, x_i', y_i\right). \tag{6}$$

The set of gradient-based class prototypes is denoted as $\mathcal{G}_k^{(t)} = \{\mathbf{g}_{k,c}^{(t)} \mid c \in C_k\}$, where $C_k \subset C$ is the set of class labels in the client dataset $\mathcal{D}_k$, and C is the set of class labels in the global dataset $\mathcal{D}$.

After local training and the extraction of gradient-based class prototypes, client k uploads the updated models $\mathbf{w}_k^{(t+1)} = (\mathbf{r}_k^{(t+1)}, \mathbf{h}_k^{(t+1)})$ along with $\mathcal{G}_k^{(t)}$ to the server for global optimization. Intuitively, $\mathcal{G}_k^{(t)}$ captures class-wise update directions, enabling the server to generate virtual features aligned with the real features in a privacy-protected manner.

Algorithm 1 The Update Procedure of Client k

Input: Global model parameters $\mathbf{w}^{(t)} = (\mathbf{r}^{(t)}, \mathbf{h}^{(t)},)$, and auxiliary classifier parameters $\mathbf{h}'^{(t)}$.
Output: Updated parameters $\mathbf{w}_k^{(t+1)} = (\mathbf{r}_k^{(t+1)}, \mathbf{h}_k^{(t+1)})$ and the set of gradient-based prototypes $\mathcal{G}_k^{(t)}$.
Procedure: Client$(\mathbf{r}^{(t)}, \mathbf{h}^{(t)}, \mathbf{h}'^{(t)})$:
Initialize local model $\mathbf{w}_k^{(t)} = (\mathbf{r}^{(t)}, \mathbf{h}^{(t)})$.
`// Local training`
for local epoch 1 to E **do**
  Update local model $\mathbf{w}_k^{(t)}$ following $\mathcal{L}_{\text{CART}}$.
end for
`// Extraction of class prototypes`
Initialize local model $\mathbf{w}'^{(t)}_k = (\mathbf{r}^{(t)}, \mathbf{h}'^{(t)})$.
for each class $c = 1$ to C_k **do**
  Extract $g_{k,c}^{(t)}$ using Eq. (6).
end for
return $\mathbf{w}_k^{(t+1)} = (\mathbf{r}_k^{(t+1)}, \mathbf{h}_k^{(t+1)})$ and $\mathcal{G}_k^{(t)}$.

4.3. Server-Side Global Optimization

On the server, the process of global optimization consists of three components: (i) global aggregation, (ii) alignment of virtual feature, and (iii) re-training of auxiliary classifier. The overall procedure of FedCART is summarized in Algorithm 2.

Global Aggregation. In round t, after receiving the local models and class prototypes from active clients, the server performs a weighted aggregation for both models and graident-based class prototypes. The aggregation of the models follows Eq. (1). Similarly, the aggregated prototype for class c is computed as:

$$\overline{\mathbf{g}}_c^{(t)} = \sum_{k \in A^{(t)}} \frac{N_k}{\sum_{k' \in A^{(t)}} N_{k'}} \mathbf{g}_{k,c}^{(t)}. \tag{7}$$

And the set of global prototypes is $\mathcal{G}^{(t)} = \{\overline{\mathbf{g}}_c^{(t)} \mid c \in \cup_k C_k,\ k \in A^{(t)}\}$.

Alignment of Virtual Feature. After obtaining the set of global prototypes $\mathcal{G}^{(t)}$, the server updates the virtual feature set $\mathcal{Z}_{\text{v}}^{(t)}$, so that the gradient-based prototypes extracted from $\mathcal{Z}_{\text{v}}^{(t)}$ for the auxiliary classifier $\mathbf{h}'^{(t)}$ are aligned with the set of aggregated class prototypes $\mathcal{G}^{(t)}$. Each class is assigned an equal number of virtual features in $\mathcal{Z}_{\text{v}}^{(t)}$, ensuring a balanced representation. Specifically, the virtual features $z_{\text{v},c}^{(t)} \in \mathcal{Z}_{\text{v}}^{(t)}$ are iteratively updated over T_V epochs with step size $\hat{\eta}_a'$ by minimizing the loss of mean squared error(MSE):

$$\mathcal{L}_{\text{MSE}}(\mathcal{Z}_{\text{v}}^{(t)}; \mathbf{h}'^{(t)}, \mathcal{G}^{(t)}) = \left\| \nabla_{\mathbf{h}'} \mathcal{L}_{\text{CE}}(\mathbf{h}'^{(t)}; \mathcal{Z}_{\text{v}}^{(t)}) - \mathcal{G}^{(t)} \right\|^2. \tag{8}$$

The virtual features are then updated as:

$$\mathcal{Z}_{\text{v}}^{(t+1)} \leftarrow \mathcal{Z}_{\text{v}}^{(t)} - \hat{\eta}_a' \nabla_z \mathcal{L}_{\text{MSE}}(\mathcal{Z}_{\text{v}}^{(t)}; \mathbf{h}'^{(t)}, \mathcal{G}^{(t)}). \tag{9}$$

Algorithm 2 Training Procedure of FedCART

Input: Initialized global model parameters $\mathbf{w}^{(0)} = (\mathbf{r}^{(0)}, \mathbf{h}^{(0)})$, auxiliary classifier parameters $\mathbf{h}'^{(0)}$, number of total communication rounds T, number of virtual feature update epochs T_V, and number of auxiliary classifier re-training epochs T_R.

Output: Trained global model $\mathbf{w}^{(T)} = (\mathbf{r}^{(T)}, \mathbf{h}'^{(T)})$.

for $t = 1$ **to** T **do**
  Randomly select active client set $A^{(t)}$.
  `// Client-side local update`
  for all each client $k \in A^{(t)}$ (in parallel) **do**
    $\{\mathbf{w}_k^{(t+1)}, \mathcal{G}_k^{(t)}\} \leftarrow \text{Client}_k(\mathbf{r}^{(t)}, \mathbf{h}^{(t)}, \mathbf{h}'^{(t)})$.
  end for
  `// Server-side global optimization`
  Aggregate $\mathbf{w}^{(t+1)}$ and $\overline{\mathbf{g}}_c^{(t)}$ using Eq. (1) and Eq. (7).
  Update virtual feature set $\mathcal{Z}_{\text{v}}^{(t)}$ using Eq. (8) and Eq. (9) for T_V epochs.
  Retrain auxiliary classifier $\mathbf{h}'^{(t)}$ using Eq. (10) for T_R epochs.
  Broadcast $(\mathbf{r}^{(t+1)}, \mathbf{h}^{(t+1)}, \mathbf{h}'^{(t+1)})$ to active clients.
end for

Re-Training of Auxiliary Classifier. Once the virtual feature set $\mathcal{Z}_{\text{v}}^{(t)}$ is updated, the auxiliary classifier $\mathbf{h}'^{(t)}$ is re-trained on it for T_R epochs with step size $\hat{\eta}_r'$ to calibrate the bias, which is updated through:

$$\mathbf{h}'^{(t+1)} \leftarrow \mathbf{h}'^{(t)} - \hat{\eta}_r' \nabla_{\mathbf{h}'} \mathcal{L}_{\text{CE}}(\mathbf{h}'^{(t)}; \mathcal{Z}_{\text{v}}^{(t)}). \quad (10)$$

The updated model $(\mathbf{r}^{(t+1)}, \mathbf{h}^{(t+1)}, \mathbf{h}'^{(t+1)})$ is then broadcast to the clients participating in the next communication round.

4.4. Privacy Analysis

As discussed in this section, FedCART involves the transmission of gradient-based prototype sets between the server and clients. However, reconstructing raw data samples from the prototypes is considered difficult [9] due to the irreversible nature of the averaging process. To further strengthen privacy protection, encryption techniques can be employed to safeguard transmitted information against possible eavesdropping by malicious adversaries.

5. Experiments

5.1. Experiment Settings

Dataset. We conduct experiments on the long-tailed variants of CIFAR-10 [13], CIFAR-100 [13], FMNIST [36], and SVHN [23]. To simulate data distributions in the real world, we partition it in two settings: (i) Long-tailed setting: following [1], we vary the imbalance ratio $\rho = n^{(1)}/n^{(|C|)}$ to control the degree of long-tailed, where a larger ρ indicates a heavier long-tailed distribution. The variant datasets are denoted as CIFAR10-LT, CIFAR100-LT, FMNIST-LT, and SVHN-LT. (ii) Non-IID setting: following [3], we use a Dirichlet distribution $p_k^c \sim Dir(\beta)$ to simulate the degree of data heterogeneity among clients, where a smaller β indicates stronger heterogeneity. Unless stated otherwise, we set $\rho = 50$ and $\beta = 0.5$ by default, and the results for different values of ρ and β are reported in Fig. 5 and Tab. 3, respectively.

Baseline. We compare FedCART with two state-of-the-art FAT methods: FedFAT [50] and CalFAT [3]. We further adapt several classical centralized adversarial training (CAT) methods, including PGD [20], TRADES [41], MART [34], and TAET [40](focuses on long-tailed distributions), to the federated setting by combining them with FedAvg [21], denoted as FedPGD, FedTRADES, FedMART, and FedTAET.

Metrics. We report natural accuracy (Natural) and robust accuracy under adversarial attacks. Specifically, we evaluate single-step attack FGSM [6] and some multi-step attacks (20 steps, step size $3/255$, $\epsilon = 8/255$) including PGD [20] and CW [2], as well as AutoAttack (AA) [5], which is an ensemble of diverse parameter-free attacks. And we report the average robust accuracy (Robust AVG) under all attacks.

Implementation. By default, we set the total communication rounds as $T = 150$, with all $K = 5$ clients participating in each round, and the number of local epochs is set to $E = 1$, consistent with the experimental setups in [3, 4]. We use PGD [20] with 10 steps to generate adversarial samples. For CIFAR10-LT and CIFAR100-LT, we adopt ResNet-18 [7] as the backbone, while a lightweight CNN architecture is used for FMNIST-LT and SVHN-LT. All methods use identical model architectures to ensure fair comparison.

5.2. Performance Comparison

As shown in Tab. 1, FedCART consistently achieves the best robust performance across all datasets, outperforming the best baseline by $1.65\% \sim 9.89\%$ on average. Under PGD-20, FedCART yields substantial gains of 2.43% on CIFAR10-LT and 6.43% on SVHN-LT. Under AA, FedCART exceeds the best baselines of 0.59% on CIFAR100-LT and 4.94% on FMNIST-LT. Although there is a trade-off between natural and robust accuracies, FedCART maintains competitive natural accuracy, trailing CalFAT by only 1.62% on FMNIST-LT while outperforming all methods in other datasets. We observe that most of methods are more fragile for AA and CW. Compared with the extension of CAT, FAT methods perform better in most cases, which implies that the direct extensions of CAT are infeasible due to the Non-IID characteristics.

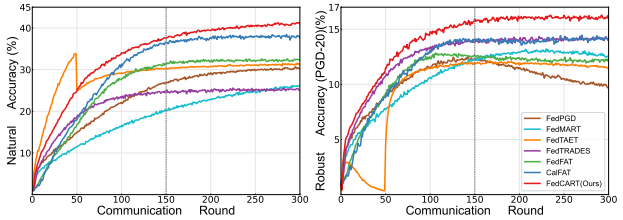
To further assess performance, we extend $T = 300$. As illustrated in Fig. 4, FedCART achieves faster convergence, which is crucial under the costly communication overhead in the FAT setting. Moreover, while methods like FedPGD exhibit robustness deterioration in the later stage of training,

Table 1. Natural and robust accuracies(%) across different datasets. Best results are in bold and the second best results are underlined. FedCART outperforms other baselines in most of cases. See details in Sec. 5.2.

Dataset	CIFAR10-LT						CIFAR100-LT					
Metric	Natural	Robust Accuracy				Robust AVG	Natural	Robust Accuracy				Robust AVG
		FGSM [6]	CW [2]	PGD-20 [20]	AA [5]			FGSM [6]	CW [2]	PGD-20 [20]	AA [5]	
FedPGD [20]	31.24	22.76	21.03	21.65	20.65	21.52	26.65	13.40	11.08	12.20	10.50	11.80
FedMART [34]	26.83	22.07	20.72	21.61	20.46	21.21	20.30	12.64	10.68	12.22	10.34	11.47
FedTAET [40]	33.98	23.62	21.49	22.42	20.80	22.08	30.41	13.22	9.84	11.85	8.91	10.96
FedTRADES [41]	41.59	26.30	23.74	25.02	23.49	24.64	24.85	14.72	12.29	14.04	11.94	13.25
FedFAT [50]	40.16	26.47	24.15	24.69	23.54	24.71	31.78	14.55	11.93	12.67	11.13	12.57
CalFAT [3]	61.25	32.86	21.08	30.04	19.93	25.98	36.86	15.85	9.18	14.00	8.40	11.86
FedCART (ours)	**66.12**	**36.59**	**30.29**	**32.47**	**27.47**	**31.71**	**38.19**	**17.37**	**14.04**	**15.65**	**12.53**	**14.90**
Dataset	SVHN-LT						FMNIST-LT					
Metric	Natural	Robust Accuracy				Robust AVG	Natural	Robust Accuracy				Robust AVG
		FGSM [6]	CW [2]	PGD-20 [20]	AA [5]			FGSM [6]	CW [2]	PGD-20 [20]	AA [5]	
FedPGD [20]	84.23	58.90	52.60	51.81	51.31	53.66	76.52	68.47	67.53	67.70	67.32	67.75
FedMART [34]	84.78	63.31	57.88	57.77	56.94	58.98	75.58	67.33	66.28	66.55	66.07	66.56
FedTAET [40]	78.30	62.70	59.07	60.98	58.52	60.32	80.21	69.34	67.66	68.16	67.30	68.11
FedTRADES [41]	77.67	63.18	59.69	60.59	59.05	60.63	80.71	68.49	66.35	66.44	65.72	66.75
FedFAT [50]	84.86	60.16	53.78	53.03	52.53	54.88	80.85	69.59	68.07	68.25	67.74	68.41
CalFAT [3]	85.93	66.59	55.63	62.86	54.72	59.95	**86.75**	73.61	65.89	71.72	65.57	69.20
FedCART (ours)	**90.98**	**74.99**	**69.73**	**69.29**	**68.06**	**70.52**	85.13	**74.75**	**72.95**	**73.64**	**72.68**	**73.51**

Table 2. Natural and robust accuracies(%) across different classes on CIFAR100-LT. Classes with more than 150 samples are the Majority. Classes with less than 45 samples are the Minority. The rest are the Medium. Best results are in bold and the second best results are underlined. See details in Sec. 5.2.

Method	Majority		Medium		Minority	
	Natural	PGD-20 [20]	Natural	PGD-20 [20]	Natural	PGD-20 [20]
FedPGD [20]	53.48	27.52	24.73	9.70	6.78	2.00
FedMART [34]	45.91	29.80	15.67	8.19	3.51	1.38
FedTAET [40]	55.41	25.19	30.82	10.27	10.22	2.50
FedTRADES [41]	53.08	**32.93**	21.27	10.31	5.15	1.93
FedFAT [50]	**60.45**	28.05	30.81	10.17	9.73	2.42
CalFAT [3]	50.17	20.70	37.35	13.61	25.90	8.99
FedCART (ours)	52.16	24.25	**39.51**	**15.34**	**26.06**	**9.09**

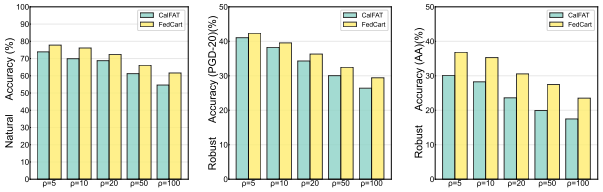


(a) (b)

Figure 4. The (a) natural accuracy and (b) robust accuracy learning curves of different methods on CIFAR100-LT. Compared to other methods, FedCART converges faster to a higher accuracy. See details in Sec. 5.2.

FedCART maintains stable and reliable performance.

As presented in Tab. 2, FedCART exhibits outstanding performance in the groups of Medium and Minority, while some other methods are nearly failing. These results under long-tailed distributions further highlight the challenges and validate the generalization of our method.



(a) (b) (c)

Figure 5. The (a) Natural accuracy, (b) Robust accuracy under PGD-20, and (c) Robust accuracy under AA of CalFAT and FedCART under different imbalance ratios ρ of CIFAR10-LT. A Larger ρ indicates a more serious long-tailed distribution. FedCART is less impaired as ρ increases. See details in Sec. 5.3.

5.3. Effect of the Degree of Long-Tailed Distribution

As illustrated in Fig. 5, we evaluate the performance of CalFAT and FedCART under varying long-tailed degrees by adjusting ρ. As ρ increases, both methods exhibit performance degradation, confirming the detrimental effect of long-tailed distributions. Nevertheless, FedCART consistently outperforms CalFAT in all settings, the performance gap widening as the long-tailed distribution intensifies. These results demonstrate FedCART's superiority in seriously long-tailed distributions.

5.4. Analysis on Data Partition

Different Degree of Imbalanced. We test the impact of different levels of data heterogeneity by varying β, as shown in Tab. 3. As data heterogeneity alleviates with increasing β, both methods exhibit improvements in natural and robust accuracies, which demonstrates that a lower degree of data heterogeneity could help the model learn. For natural accuracy, FedCART outperforms CalFAT across almost all settings except when $\beta = 0.25$. For robust accuracy, Fed-

Table 3. Natural and robust accuracies(%) across different β on CIFAR10-LT. FedCART outperforms CalFAT in most settings. Best results are in bold. See details in Sec. 5.4.

Non-IID	Method	Natural	PGD-20 [20]	AA [5]
$\beta = 0.25$	CalFAT [3]	**55.65**	27.35	18.00
	FedCART	51.52	**30.58**	**24.72**
$\beta = 0.5$	CalFAT [3]	61.25	30.04	19.93
	FedCART	**66.12**	**32.47**	**27.47**
$\beta = 0.75$	CalFAT [3]	63.25	30.22	20.06
	FedCART	**69.29**	**33.16**	**29.28**

CART performs better than CalFAT, especially under AA attack, which demonstrates superior stability of robustness under various Non-IID scenarios.

Table 4. Natural and robust accuracies(%) across different number of clients and active rates on CIFAR10-LT. FedCART performs better in most of settings. Best results are in bold. See details in Sec. 5.4.

client number		$K = 5$			$K = 20$		
active rates	Method	Natural	PGD-20 [20]	AA [5]	Natural	PGD-20 [20]	AA [5]
0.1	CalFAT [3]	–	–	–	33.71	20.13	12.89
	FedCART	–	–	–	**34.34**	**21.70**	**19.10**
0.2	CalFAT [3]	38.11	19.59	13.37	30.75	19.57	11.92
	FedCART	**44.38**	**21.46**	**18.77**	**42.95**	**25.04**	**20.85**
0.5	CalFAT [3]	**53.48**	26.83	17.46	40.04	21.04	9.92
	FedCART	51.41	**27.22**	**23.25**	**43.38**	**23.21**	**18.23**
0.7	CalFAT [3]	54.10	26.32	17.68	35.94	20.59	10.85
	FedCART	**63.00**	**27.32**	**24.10**	**46.33**	**26.05**	**21.42**
1.0	CalFAT [3]	61.30	31.30	21.05	43.32	20.99	9.98
	FedCART	**67.24**	**32.57**	**27.90**	**47.06**	**27.18**	**22.75**

Different Number of Clients. We compare FedCART with CalFAT under varying client numbers and active rates, results shown in Tab. 4. Performance degrades as the number of clients increases or the rate of participation decreases in general, due to the lack of data for clients in each communication round. Across nearly all configurations, FedCART performs better in natural accuracy, and FedCART achieves the highest robust accuracy under PGD-20 and AA in all settings. These results validate the reliability of FedCART in various settings of client participation.

5.5. Contribution of Each Component in FedCART

Table 5. The accuracy(%) of ablation study for key components of FedCART on CIFAR10-LT. Best results are in bold. The results show that each component plays an important role in FedCART. See details in Sec. 5.5.

Case	Configuration					Natural	PGD-20 [20]	AA [5]
	$\mathcal{L}_{NAT}$	$\mathcal{L}_{AT}$	$\mathcal{L}_{Align}$	CART	$\mathbf{h}'$			
1	✓			✓	✓	**79.08**	0.00	0.00
2		✓		✓	✓	53.86	29.79	26.72
3	✓		✓	✓	✓	75.30	2.91	2.22
4		✓	✓	✓	✓	49.91	32.54	26.98
5	✓	✓		✓	✓	66.04	29.33	26.12
6	✓	✓	✓			40.95	24.58	23.48
7	✓	✓	✓	✓		49.98	28.30	25.69
Ours (Full)	✓	✓	✓	✓	✓	67.24	**32.57**	**27.90**

As described in Sec. 4, FedCART consists of client-side local update and server-side global optimization, which includes the re-training process(CART) for the auxiliary classifier $\mathbf{h}'$. Local loss $\mathcal{L}_{\text{CART}}$ combines natural loss $\mathcal{L}_{\text{NAT}}$, adversarial training loss $\mathcal{L}_{\text{AT}}$, and alignment loss $\mathcal{L}_{\text{Align}}$. We explore the effectiveness of each component in FedCART.

The ablation results shown in Tab. 5 reveal a trade-off between natural and robust accuracies. Cases 1 and 3 achieve high natural accuracy (79.08% and 75.30%) but fail under adversarial attacks because of lack of $\mathcal{L}_{\text{AT}}$. Case 4 attains near-maximum robust accuracy, though with the sacrifice of natural accuracy. Case 5 closely matches the full model, but the performance gap highlights the effectiveness of the alignment loss $\mathcal{L}_{\text{Align}}$. Cases 6 and 7 show the contribution of CART and the auxiliary classifier $\mathbf{h}'$ is evident, which demonstrates that removing either leads to substantial performance drops, underscoring their role. Overall, these results validate that each component of FedCART is essential for building a FAT model that effectively balances robustness and accuracy under long-tailed distributions.

5.6. Combining FedCART with other methods

Table 6. The natural and robust accuracies(%) of the combination of our framework with different CAT methods on CIFAR10-LT.

Metric	Natural	PGD-20
FedMART	26.83	21.61
MART + FedCART (ours)	33.05	27.04
FedTAET	33.98	22.42
TAET + FedCART (ours)	46.61	26.57
FedTRADES	41.59	25.02
TRADES + FedCART (ours)	65.48	27.93
FedCART (ours)	66.12	32.47

To further validate the effectiveness of our framework, we combine it with three other CAT methods. As shown in Tab. 6, combining these methods with FedCART improves both natural and robust accuracies compared to directly extending them with FedAvg [21]. These results confirm the effectiveness and extensibility of our framework, offering an alternative approach to adapt CAT to the FAT setting.

6. Conclusion

In this paper, we study the problem of adversarial robustness in FL under long-tailed distributions. To address this challenge, we propose FedCART, a novel FAT framework which includes a client-side alignment loss with a server-side calibration mechanism for the classifier by using class-balanced virtual features. Extensive experiments in diverse settings show that FedCART outperforms state-of-the-art baselines in both natural and robust accuracies. In future work, we plan to further extend this framework to address additional security concerns. To the best of our knowledge, this is the first work to investigate FAT under long-tailed distributions. We believe that it will be an important step toward the adversarial robustness of FL in the real world.

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